

SAMIM REZA

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EDUCATION

Green University of Bangladesh B.Sc. in Computer Science and Engineering (CGPA: 3.80/4.00)	2022 – 2025
Government Bangabandhu College Higher Secondary Certificate (GPA: 4.75/5.00)	2017 – 2019
S. M. Model Govt. High School Secondary School Certificate (GPA: 5.00/5.00)	2012 – 2017

WORK EXPERIENCE

Lead Engineer — <i>SolaneCode, New York</i> • Lead engineering for a live restaurant AI phone receptionist SaaS sold in New York • Built and own the AI voice workflow, frontend, backend, QA, CRM, menu, kitchen flow, and reporting systems	[Live Platform] May 2026 – Present
Machine Learning Engineer — <i>Brandifies</i> • Built end-to-end ML pipelines, Fine-tuned LLMs and implemented RAG-based chatbot systems • Developed predictive models using boosting algorithms	Dec 2025 – Present
Robotics Engineer Intern — <i>RoboTech Valley</i> • Designed robotics modules using embedded systems and ROS for autonomous navigation	Jul – Sept 2025
Teaching Assistant — <i>Green University of Bangladesh</i> • Conducted demo classes on C Programming and Data Structures • Evaluated exam scripts and prepared grade sheets	May – Aug 2025
Programming Trainer — <i>Green University of Bangladesh</i> • Trained students for competitive programming through weekly problem-solving sessions	Jan – Dec 2025
Mentor — <i>University Mentorship Program</i> • Mentored first-year students in programming fundamentals and professional tools	Jun 2023 – Aug 2025

RESEARCH

- **Edge Detection for Smoke-Degraded Images using Fractional Derivative** (Published, 2026) – Proposed a fractional calculus-based edge detection operator achieving robust boundary extraction under smoke and haze degradation conditions
- **Denoising Diffusion Probabilistic Models** – Integrating edge detection for enhanced image generation
- **Federated Learning** – Adaptive client selection algorithms for optimal convergence

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, VectorDB, RAG, LLM, Roboflow

Voice AI: OpenAI Realtime, Twilio Media Streams, tool-calling agents, Chroma-backed RAG

Web/Backend: Django, FastAPI, Next.js, React, REST APIs, PostgreSQL, MySQL, PWA

Tools & Platforms: Git, Docker, Celery, Redis, Twilio, Stripe, Meta Graph API, Linux, LaTeX, Supabase, Render, Arduino, Raspberry Pi, Jetson, ESP32, Azure, Digital Ocean

FEATURED PROJECTS

- Personal AI Assistant** — LLM, RAG, LangChain [Live] | [GitHub]
• Built an LLM-powered chatbot with retrieval-augmented generation for personalized assistance, deployed at samimreza.me/chat
- TrueDoc** — Django, React, DRF, SimpleJWT, PostgreSQL, QR [GitHub]
• SaaS for digital document attestation in Bangladesh with QR-backed verification, role-based access (citizen/verifier/org), JWT auth, and BDT payment model
- GoUp** — Django, Celery, Redis, Meta Graph API, Twilio [GitHub]
• For Lead Ads automation; syncs pages/forms via Meta API, webhooks, triggers SMS follow-ups Celery & Redis
- KobitaLM** — Bengali Poetry Generator [Model] | [GitHub]
• Fine-tuned Gemma 2 9B with LoRA and RAG for Bengali poetry generation
- WeCare Medical Service App** — FastAPI, Qwen3-VL, PWA [GitHub]
• Developed an offline-first PWA for rural healthcare with AI-assisted triage capabilities
- Cirrhosis Outcome Prediction** — XGBoost, LightGBM, CatBoost BAIC 2025
• Built an ensemble ML classifier with 55 engineered features, achieving 14.5% accuracy gain
- Hasib Academic System** — Django, PostgreSQL, Render, Unicorn [Live] | [GitHub]
• A production-ready Django application for school workflows including admissions, student attendance, notices, teacher accounts, and finance tracking with PDF export. Tailored for Bangla-speaking users
- Union Ledger** — Django, PostgreSQL, Render [Live] | [GitHub]
• Built a collaborative financial tracking web app with consensus-based voting for deposits, loans, repayments, and investments
- Mars Rover** — ROS, Embedded Systems [GitHub]
• Developed a ROS-based autonomous navigation system for Mars rover simulation

COMPETITIVE PROGRAMMING

- **Champion:** Intra Dept Programming Contest 2024
- **3x ICPC Regionalist:** 2022, 2023, 2024
- **Kaggle Competitor:** Active in ML competitions
- **Problem Setter:** Junior Programming Contests
- **Participated:** NCPC 2023, IUPC 2023/2024

ACHIEVEMENTS

- Vice Chancellor's Award for Academic Excellence
- Champion, Inter Dept Programming Contest 2024
- Secured **8th** position in BUBT Artificial Intelligence Conquest in Kaggle
- National IT Quiz Competition Medalist

REFERENCES

Prof. Dr. Md. Saiful Azad
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